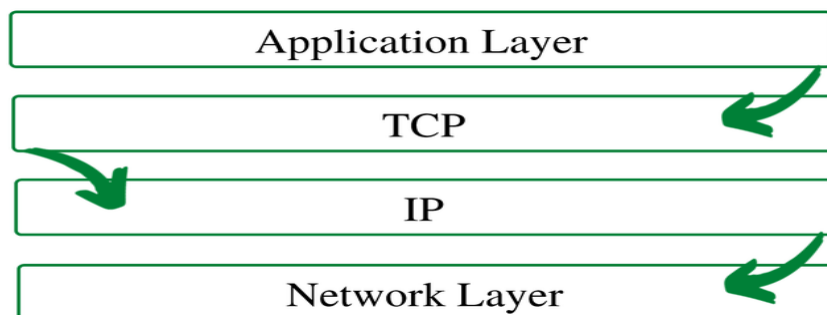


Unit-4 - NOTES

TCP (Transmission Control Protocol)

TCP is one of the main protocols of the TCP/IP suite. It lies between the Application and Network Layers which are used in providing reliable delivery services. Transmission Control Protocol (TCP) ensures reliable and efficient data transmission over the internet. TCP plays a crucial role in managing the flow of data between computers, guaranteeing that information is delivered accurately and in the correct sequence.

Transmission Control Protocol (TCP) is a connection-oriented protocol for communications that helps in the exchange of messages between different devices over a network. The Internet Protocol (IP), which establishes the technique for sending data packets between computers, works with TCP.



Internet Protocol (IP)

Internet Protocol (IP) is a method that is useful for sending data from one device to another from all over the internet. It is a set of rules governing how data is sent and received over the internet. It is responsible for addressing and routing packets of data so they can travel from the sender to the correct destination across multiple networks. Every device contains a unique IP Address that helps it communicate and exchange data across other devices present on the internet.

Features of TCP/IP

- **Segment Numbering System:** TCP keeps track of the segments being transmitted or received by assigning numbers to each and every single one of them. A specific Byte Number is assigned to data bytes that are to be transferred while segments are assigned *sequence numbers*. Acknowledgment Numbers are assigned to received segments.
- **Connection Oriented:** It means sender and receiver are connected to each other till the completion of the process. The order of the data is maintained i.e. order remains same before and after transmission.
- **Full Duplex:** In TCP data can be transmitted from receiver to the sender or vice – versa at the same time. It increases efficiency of data flow between sender and receiver.
- **Flow Control:** Flow control limits the rate at which a sender transfers data. This is done to ensure reliable delivery. The receiver continually hints to the sender on how much data can be received (using a sliding window).
- **Error Control:** TCP implements an error control mechanism for reliable data transfer. Error control is byte-oriented. Segments are checked for error detection. Error Control includes – Corrupted Segment & Lost Segment Management, Out-of- order segments, Duplicate segments, etc.
- **Congestion Control:** TCP takes into account the level of congestion in the network. Congestion level is determined by the amount of data sent by a sender.

Advantages of TCP

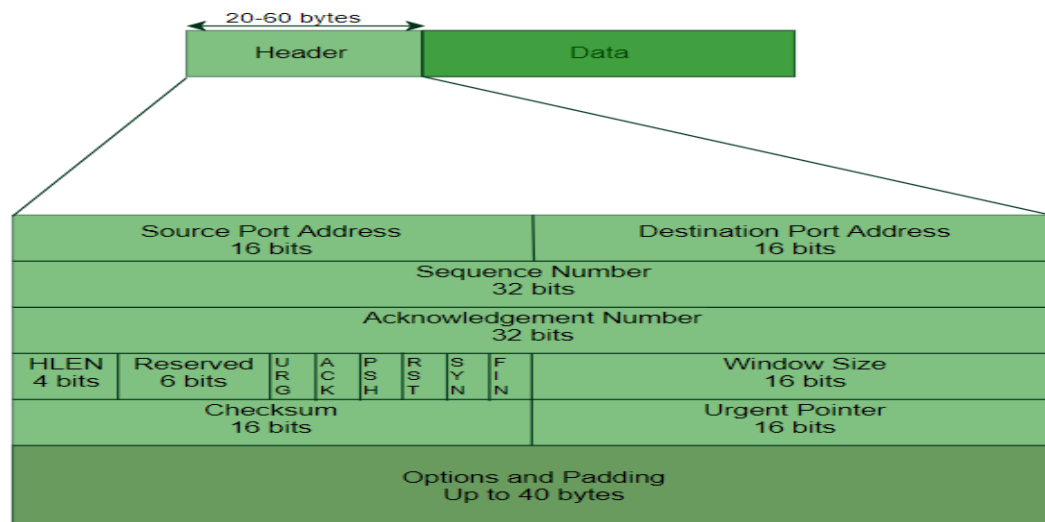
- It is a reliable protocol.
- It provides an error-checking mechanism as well as one for recovery.
- It gives flow control.
- It makes sure that the data reaches the proper destination in the exact order that it was sent.

- It is a well-documented and widely implemented protocol, maintained by standards organizations like the IETF (Internet Engineering Task Force).
- It works in conjunction with IP (Internet Protocol) to establish connections between devices on a network.

Disadvantages of TCP

- TCP is made for Wide Area Networks, thus its size can become an issue for small networks with low resources.
- TCP runs several layers so it can slow down the speed of the network.

TCP Header



The header of a TCP segment can range from 20-60 bytes. 40 bytes are for options. If there are no options, a header is 20 bytes else it can be of upmost 60 bytes. Header fields:

- **Source Port Address –**
A 16-bit field that holds the port address of the application that is sending the data segment.
- **Destination Port Address –**
A 16-bit field that holds the port address of the application in the host that is receiving the data segment.
- **Sequence Number –**
A 32-bit field that holds the sequence number, i.e, the byte number of the first byte that is sent in that particular segment. It is used to reassemble the message at the receiving end of the segments that are received out of order.

- **Acknowledgement Number –**

A 32-bit field that holds the acknowledgement number, i.e, the byte number that the receiver expects to receive next. It is an acknowledgement for the previous bytes being received successfully.

- **Header Length (HLEN) –**

This is a 4-bit field that indicates the length of the TCP header by a number of 4-byte words in the header, i.e if the header is 20 bytes(min length of TCP header), then this field will hold 5 (because $5 \times 4 = 20$) and the maximum length: 60 bytes, then it'll hold the value 15(because $15 \times 4 = 60$). Hence, the value of this field is always between 5 and 15.

- **Control flags –**

These are 6 1-bit control bits that control connection establishment, connection termination, connection abortion, flow control, mode of transfer etc. Their function is:

- URG: Urgent pointer is valid
- ACK: Acknowledgement number is valid(used in case of cumulative acknowledgement)
- PSH: Request for push
- RST: Reset the connection
- SYN: Synchronize sequence numbers
- FIN: Terminate the connection

- **Window size –**

This field tells the window size of the sending TCP in bytes.

- **Checksum –**

This field holds the checksum for error control. It is mandatory in TCP as opposed to UDP.

- **Urgent pointer –**

This field (valid only if the URG control flag is set) is used to point to data that is urgently required that needs to reach the receiving process at the earliest. The value of this field is added to the sequence number to get the byte number of the last urgent byte.

User Datagram Protocol (UDP)

UDP is a Transport Layer protocol. UDP is a part of the Internet Protocol suite, referred to as UDP/IP suite. Unlike TCP, it is an unreliable and connectionless protocol. So, there is no need to establish a connection before data transfer. The UDP helps to establish low-latency and loss-tolerating connections over the network. The UDP enables process-to-process communication.

Unlike Transmission Control Protocol (TCP), UDP is connectionless and does not guarantee delivery, order, or error checking, making it a lightweight and efficient option for certain types of data transmission.

UDP Header

UDP header is an 8-byte fixed and simple header, while for TCP it may vary from 20 bytes to 60 bytes. The first 8 Bytes contain all necessary header information and the remaining part consists of data. UDP port number fields are each 16 bits long, therefore the range for port numbers is defined from 0 to 65535; port number 0 is reserved.



UDP Header

- Source Port: Source Port is a 2 Byte long field used to identify the port number of the source.
- Destination Port: It is a 2 Byte long field, used to identify the port of

the destined packet.

- Length: Length is the length of UDP including the header and the data. It is a 16-bits field.
- Checksum: Checksum is 2 Bytes long field.

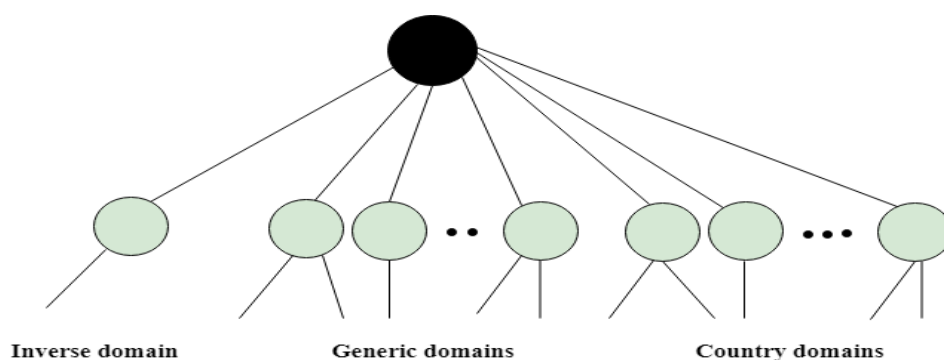
Applications of UDP

- Used for simple request-response communication when the size of data is less and hence there is lesser concern about flow and error control.
- It is a suitable protocol for multicasting as UDP supports packet switching.
- UDP is used for some routing update protocols like RIP(Routing Information Protocol).
- VoIP (Voice over Internet Protocol) services, such as Skype and WhatsApp, use UDP for real-time voice communication.
- DNS (Domain Name System) also uses UDP for its query/response messages. DNS queries are typically small and require a quick response time, making UDP a suitable protocol for this application.
- DHCP (Dynamic Host Configuration Protocol) uses UDP to dynamically assign IP addresses to devices on a network. DHCP messages are typically small, and the delay caused by packet loss or retransmission is generally not critical for this application.

DNS (Domain Name System)

- DNS is a directory service that provides a mapping between the name of a host on the network and its numerical address.
- DNS is required for the functioning of the internet.
- Each node in a tree has a domain name, and a full domain name is a sequence of symbols specified by dots.
- DNS is a service that translates the domain name into IP addresses. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.
- For example, suppose the FTP site at EduSoft had an IP address of 132.147.165.50, most people would reach this site by specifying ftp.EduSoft.com. Therefore, the domain name is more reliable than IP address.

DNS is a TCP/IP protocol used on different platforms. The domain name space is divided into three different sections: generic domains, country domains, and inverse domain.



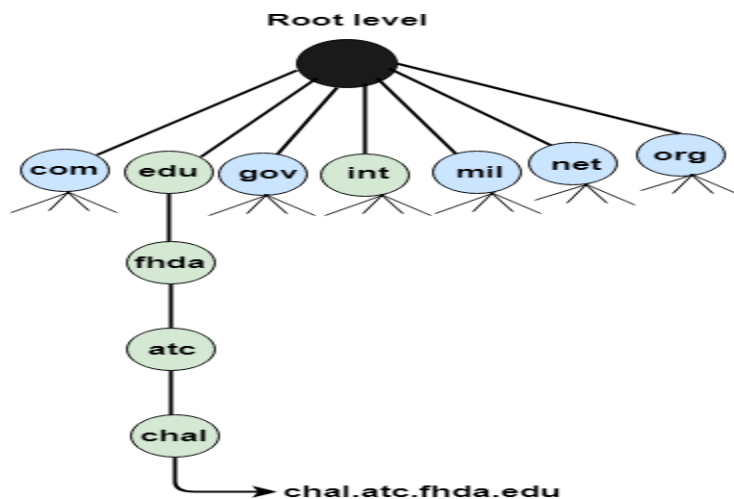
Generic Domains

- It defines the registered hosts according to their generic behavior.
- Each node in a tree defines the domain name, which is an index to the DNS database.

- It uses three-character labels, and these labels describe the organization type.

Generic domain labels

<i>Label</i>	<i>Description</i>
aero	Airlines and aerospace companies
biz	Businesses or firms (similar to “com”)
com	Commercial organizations
coop	Cooperative business organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International organizations
mil	Military groups
museum	Museums and other nonprofit organizations
name	Personal names (individuals)
net	Network support centers
org	Nonprofit organizations
pro	Professional individual organizations



Country Domain

The format of country domain is same as a generic domain, but it uses two-character country abbreviations (e.g., us for the United States) in place of three character organizational abbreviations.

Inverse Domain

The inverse domain is used for mapping an address to a name. When the server has received a request from the client, and the server contains the files of only authorized clients. To determine whether the client is on the authorized list or not, it sends a query to the DNS server and ask for mapping an address to the name.

Working of DNS

- DNS is a client/server network communication protocol. DNS clients send requests to the server while DNS servers send responses to the client.
- Client requests contain a name which is converted into an IP address known as a forward DNS lookup, while requests containing an IP address which is converted into a name known as reverse DNS lookups.
- DNS implements a distributed database to store the name of all the hosts available on the internet.
- If a client like a web browser sends a request containing a hostname, then a piece of software such as DNS resolver sends a request to the DNS server to obtain the IP address of a hostname. If DNS server does not contain the IP address associated with a hostname, then it forwards the request to another DNS server. If IP address has arrived at the resolver, which in turn completes the request over the internet protocol.

Audio and Video Compression

Audio/Video compression is the process of reducing the file size of an audio file while attempting to preserve as much of its quality as possible. This can be useful for saving storage space or for optimizing audio for streaming or transmission. There are two main types of audio compression:

1. Lossy Compression:

- **Description:** Lossy compression reduces file size by permanently removing audio data that is considered less audible to the human ear. This results in some loss of quality, but it can achieve much higher compression ratios.
- **Common Formats:**
 - **MP3 (MPEG Audio Layer III):** One of the most popular lossy formats, widely used for music, podcasts, and other audio content. MP3 files typically use bit rates between 128 kbps (kilobits per second) and 320 kbps.
 - **AAC (Advanced Audio Coding):** Known for providing better audio quality at similar bit rates compared to MP3. It is used by platforms like Apple Music and YouTube.
 - **Ogg Vorbis:** An open-source codec known for providing high-quality audio at lower bit rates. Often used in gaming and streaming applications.
 - **Opus:** A codec designed for both speech and music with superior quality at lower bit rates. It's used in real-time applications like VoIP (Voice over IP) and streaming.

2. Lossless Compression:

- **Description:** Lossless compression reduces the file size without losing any audio data, meaning the original quality is perfectly preserved. The file size reduction is not as significant as with lossy compression but is still

useful for archiving or when audio fidelity is a priority.

- **Common Formats:**

- **FLAC (Free Lossless Audio Codec):** A popular lossless format that reduces file sizes significantly without losing any quality. It is widely supported by media players and streaming platforms.
- **ALAC (Apple Lossless Audio Codec):** Apple's lossless format, similar to FLAC, but used primarily in Apple ecosystems.
- **WAV (Waveform Audio File Format):** A standard for uncompressed audio files. While it doesn't provide compression, it is often used for professional audio editing and production.
- **AIFF (Audio Interchange File Format):** Similar to WAV, AIFF is an uncompressed format used primarily by Apple systems.

Key Concepts in Audio Compression:

- **Bitrate:** Refers to the amount of data processed per unit of time, usually measured in kilobits per second (kbps). Higher bitrates generally lead to better quality audio but larger file sizes.
- **Sample Rate:** The number of samples of audio taken per second. Common sample rates include 44.1 kHz (standard for CD-quality audio) and 48 kHz (used in professional video).
- **Bit Depth:** Refers to the number of bits used to represent each audio sample. Higher bit depths result in higher dynamic range (the difference between the quietest and loudest parts of the sound), but also larger file sizes.

MPEG Video Compression

MPEG video compression is based on two key concepts:

- **Spatial Compression** (Intra-frame): This reduces redundancies within a single frame (image).
- **Temporal Compression** (Inter-frame): This reduces redundancies between consecutive frames.

Key Steps in MPEG Video Compression:

1. Pre-Processing:

- The video frame is prepared, including tasks like **color space conversion** (e.g., from RGB to YCbCr) and downscaling (if needed).

2. Motion Estimation:

- The encoder analyzes the video and identifies areas where pixel movement occurs between frames. This information is stored as **motion vectors**, which describe how parts of the frame have moved relative to other frames.

3. Motion Compensation:

- Using the motion vectors, the encoder predicts parts of the current frame by referring to previous frames, which helps avoid encoding the same information repeatedly.

4. Intra-frame Coding (Spatial Compression):

- For **I-frames** (Intra-coded frames), the encoder compresses each frame independently, using techniques like **Discrete Cosine Transform (DCT)** and **Quantization**. This step reduces spatial redundancies within each individual frame.

5. **Inter-frame Coding (Temporal Compression):**

- **P-frames** (Predicted frames) and **B-frames** (Bidirectionally predicted frames) are encoded using the motion vectors calculated earlier. These frames store differences between the predicted frame and the actual frame.
- **P-frames** use information from previous frames to predict the current frame.
- **B-frames** use information from both previous and subsequent frames to make predictions, resulting in even greater compression.

6. **Quantization:**

- Quantization reduces the precision of the transformed coefficients (e.g., DCT coefficients). This step introduces some loss in quality but is essential for reducing file size.

7. **Entropy Coding:**

- After quantization, **entropy coding** techniques such as **Huffman coding** or **Arithmetic coding** are applied to represent the remaining data more efficiently by using variable-length codes for more frequent values.

8. **Multiplexing:**

- The final step involves **multiplexing** the encoded video with audio data into a single file or stream (e.g., an MPEG-2 or MPEG-4 file format). This allows for synchronized playback of video and audio.

MPEG Audio Compression

MPEG audio compression is designed to reduce the size of audio files while preserving sound quality. It uses **psychoacoustic models** to remove sounds that are inaudible or less noticeable to the human ear.

Key Steps in MPEG Audio Compression:

1. Pre-Processing:

- Audio input is typically **converted to a frequency domain** (e.g., using **Fourier transform** or **MDCT – Modified Discrete Cosine Transform**) to make it easier to apply compression techniques.

2. Psychoacoustic Model:

- This model uses knowledge of human hearing to remove inaudible or redundant parts of the audio signal. For example, sounds masked by louder sounds or outside the range of human hearing are discarded. This is known as **perceptual coding**.

3. Quantization:

- The frequency-domain coefficients are quantized, meaning they are rounded to fewer bits. This step introduces **loss** but significantly reduces data size.

4. Huffman Coding:

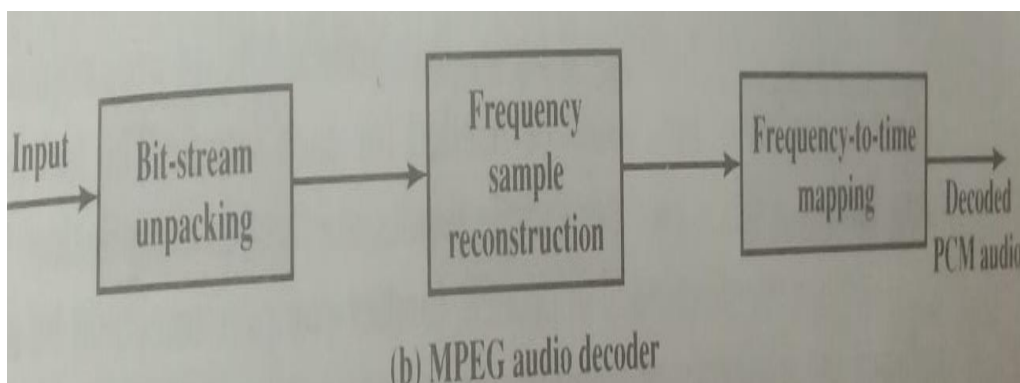
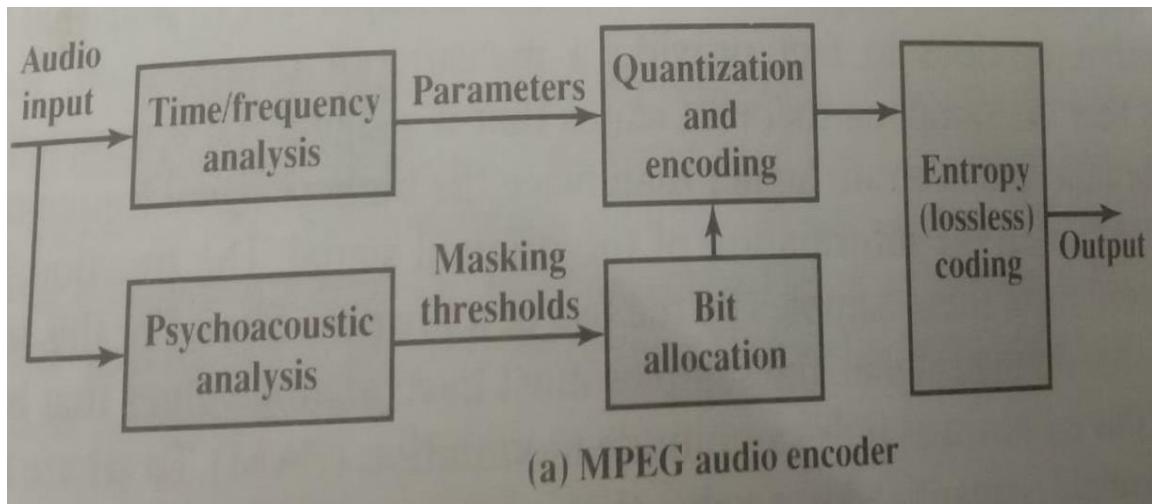
- This is an entropy encoding technique where more frequently occurring audio patterns are assigned shorter codes, further reducing the size of the audio data.

5. Bitrate Control:

- The encoder adjusts the amount of data used for encoding depending on the desired output quality and target bitrate. Higher bitrates generally result in better quality audio.

6. Multiplexing:

- The compressed audio data is multiplexed with the video stream into a single MPEG container (e.g., MPEG-2 or MPEG-4), allowing synchronized playback of video and audio.



Comparison of MPEG Video and Audio Compression:

Aspect	MPEG Video Compression	MPEG Audio Compression
Compression Type	Temporal and Spatial Compression (Lossy)	Perceptual Coding (Lossy)
Compression Techniques	Motion Estimation, DCT, Quantization, Huffman Coding	Psychoacoustic Model, Quantization, Huffman Coding

File Size	Can vary significantly depending on resolution, bitrate, and codec (MPEG-2, MPEG-4, etc.)	Can vary with bitrate and codec (MP3, AAC, etc.)
Common Formats	MPEG-1, MPEG-2, MPEG-4 (H.264, HEVC)	MP3, AAC, HE-AAC, AC-3
Applications	Video streaming, DVD/Blu-ray, Broadcast, Web video	Music, Podcasts, Streaming, Telephony (VoIP)

Multimedia Applications - SIP and VoIP

SIP and **VoIP** are utilized for data transfer over the internet, although they are different.

VoIP is independently utilized for IP telephony, whereas SIP protocol manages the complete exchange of multimedia. The SIP signalling protocol is utilized to standardize IP telephony or VoIP. It is utilized to make internet phone calls, video calls, and other multimedia connections. In contrast, **Voice over IP (VoIP)** is utilized to transport voice communication across data networks.

SIP is an abbreviation for "**Session Initiation Protocol**". It is a collection of rules governing multimedia exchange over the internet. The application layer control protocol is responsible for establishing, controlling, and terminating multimedia sessions with one or multiple users. The multimedia session may include any type of multimedia, such as data, sound, video, picture, etc. In a nutshell, a SIP function is in the same way as HTTP does, using a request and response model.

SIP also supports trunking, which allows you to connect to a traditional phone network, like a landline or mobile device. It is useful since it permits you to call any phone number, whether it is connected to the internet or not. It supports a more flexible and user-friendly phone system when integrated into a VoIP system.

How does the SIP work?

There are some steps:

Firstly, the SIP caller makes a request and sends an invite message to the receiver.

1. After that, the proxy server examines the message body structure, which includes media type, format, and the caller's capabilities.
2. If the callee approves the request, the caller receives the response code.

The callee may also use the OPTIONS mechanism to inquire about the

host's capabilities and other information.

3. The connection is finished utilizing the three-way handshaking protocol.
 4. After that, the caller sends an ACK message to end the protocol and confirm receipt of the 200 (OK) messages.
 5. In the end, the session is ended by sending a BYE method to any party.
- Component of SIP Protocol

There are mainly four types of SIP components. These components are as follows:

1. **User Agents**

The client and server are both user agents, where the client makes the requests, and the server receives and responds to them.

2. **Several Servers**

The SIP protocol uses various servers, including proxy, location, registrar, and redirect servers. Each server works on criteria.

3. **Gateways**

A gateway is just a user agent that can connect to other networks, like the PSTN.

4. **B2B(Business-to-Business) User Agents**

It includes two user agents that may transmit and alter SIP messages.

Advantages and Disadvantages of SIP

Advantages

1. It may support numerous users with different skills. For example, in a conference with some users who have video capability and others who only have audio capabilities, the video users may see each other. They are not required to use exclusively audio, as with other protocols.
2. It operates independently of the session type or medium used and provides flexibility.
3. Its messages are in plain text, which makes troubleshooting easier.
4. It is an open standard that supports and integrates several vendors.

Disadvantages

1. Fewer people are familiar with this protocol because it is a relatively new protocol. If you want to use SIP protocol within your network, make sure you have trained support staff.
2. SIP features are continually being developed, although many companies have their own proprietary protocol implementations.
3. Text message processing places a greater strain on gateways. The text must be translated into a language that the router understands.

VoIP is an abbreviation for "Voice over Internet Protocol"

VoIP technology enables voice calls to be made utilizing a broadband Internet connection rather than a traditional phone line. Some VoIP service or device enables you to call many persons simultaneously, while others could enable you to call anyone. They may or may not have a contact number, including long-distance, mobile, local, and international lines. While some VoIP services require you to utilize a particular VoIP phone or your computer, others let you utilize a standard phone connected to a VoIP adaptor.

Working of VoIP

VoIP services turn your voice into a digital signal for Internet transmission. When you call a regular telephone number, the signal is changed into a typical telephone signal before reaching the intended destination. It allows you to place calls from a system, a dedicated VoIP-specific phone, or a regular phone that is linked to an adaptor. Additionally, wireless "**hot spots**" in public spaces like parks, airports, and cafes let you access the internet and may use VoIP services wirelessly.

Advantages and Disadvantages of VoIP

Advantages

1. Some VoIP providers include capabilities and services that are either not offered by conventional phones or are offered but at a higher cost.
2. The connection can be offered more smoothly than with an analog signal.
3. It is possible to avoid paying for both an internet connection and a landline.

Disadvantages

1. Not all VoIP devices utilize emergency service contact numbers to connect to emergency services.
2. Some VoIP devices cannot work properly during power outages, and the service provider may not supply backup power.

3. VoIP service providers can or cannot offer directory assistance.

Key differences between the SIP and VoIP

1. SIP is an abbreviation for Session Initiation Protocol. In contrast, VoIP is an abbreviation for Voice over Internet Protocol.
2. SIP is a set of rules governing multimedia exchange over the internet. In contrast, VoIP technology enables voice calls to be made utilizing a broadband Internet connection rather than a traditional phone line.
3. SIP is more flexible than VoIP. In contrast, VoIP is less flexible than the SIP protocol.
4. SIP is an application layer protocol. In contrast, VoIP is a **Layer 3** network protocol.
5. SIP systems operate independently of other devices and just need a modem to work. In contrast, VoIP systems require a PC to make and receive calls.
6. SIP protocol may transport any form of media. On the other hand, VoIP protocol may only send and receive voice calls and messages.
7. Both SIP and VoIP protocols are compatible with the internet.
8. The SIP protocol doesn't depend on any other device. In contrast, VoIP protocol depends on the devices that provide internet.
9. The SIP protocol is connected to a signalling protocol to manipulate technologies. In contrast, VoIP protocol is associated with distinct and widely recognized technology.
10. SIP utilizes a number of systems to handle various functions. In contrast, the VoIP protocol utilizes a single system to handle all functions.